

Winter 2026

ENERGY295-Electrochemical Energy Storage Systems: Modeling and Estimation

Lecture 11

SIMONA ONORI, Stanford University

Energy Science and Engineering

Electrical Engineering (courtesy)

Precourt Institute of Energy



solving energy
for humanity.

Stanford | Doerr
School of Sustainability
Energy Science & Engineering

Housekeeping item

- February 23
 - 10 min Quiz 4
 - Guest Lecture

The Fragmented Battery Ecosystem

In many medium-to-large organizations, battery-related work is split into:

- **Cell Engineering / Electrochemistry Group**

Physics, DFN, degradation, materials.

- **BMS / Controls Group**

ECM, SOC/SOH estimation, embedded systems.

- **ML / Data Science Group**

Data-driven models, predictive analytics, anomaly detection.

Historically, these three groups operate with limited overlap.

From Physics-Based Modeling to SOC Estimation and BMS

- We are moving **from understanding the battery** (i.e. SPM) **to estimating and managing the battery in real time** (i.e. BMS) which requires new mathematics (dynamic systems & control)

In a BMS, the goal is:

- Estimating internal states in real time
- Operating under measurement noise
- Working with partial observability
- Implementing algorithms on embedded hardware
- Making decisions under uncertainty

SOC Estimation problem

- We only measure current and voltage, the problem becomes:
How do we reconstruct hidden internal states from limited measurements?
- **This is an estimation problem, not just a modeling problem (SPM alone is not enough).**
- SOC is not directly measurable in a battery, so it must be inferred from other measurements or estimated in some way
- From this information, the SOC estimate is constructed in real time and provided to the user by using a model (i.e., model-based estimation)

New mathematical tools we now need

To move toward SOC estimation and BMS design, we introduced:

1. State-Space Representation

Rewriting battery models as:

$$\begin{aligned}\dot{x} &= f(x, u) \\ y &= h(x, u)\end{aligned}$$

where: x = internal states (e.g., SOC, concentrations); u = current input; y = measured voltage

In the next two lectures, we will cover:

2. Observability Analysis

- Can SOC be reconstructed from voltage and current? If so, how well?
- Under what conditions?

3. Observers and Filters (open loop versus closed loop)

- Luenberger observers
- Extended Kalman Filters
- Unscented Kalman Filters
- Nonlinear observers

What is also needed that we don't cover in ENERGY295

Parameter Identification

- ECM parameter fitting
- Online adaptation
- Aging-aware estimation

Control-Oriented Model Reduction

- DFN $\rightarrow$ SPM $\rightarrow$ ECM
- Preserve physical meaning
- Ensure real-time feasibility

Data-Driven & AI/ML Methods

- Forecasting (SOC, SOH, RUL)
- Fleet-level learning
- Anomaly detection
- Physics-informed machine learning

Battery Management System (BMS)

sensors

From

- Current
- Voltage
- Temperature

measurements

BMS



functionalities

Implements critical operations

Safety Operation and Diagnosis

- Over charging
- Over heating
- Over discharging
- Short circuit

Monitoring

- State of Charge (SOC)/State of Energy
- State of Health (SOH)
- Temperature
- Voltage

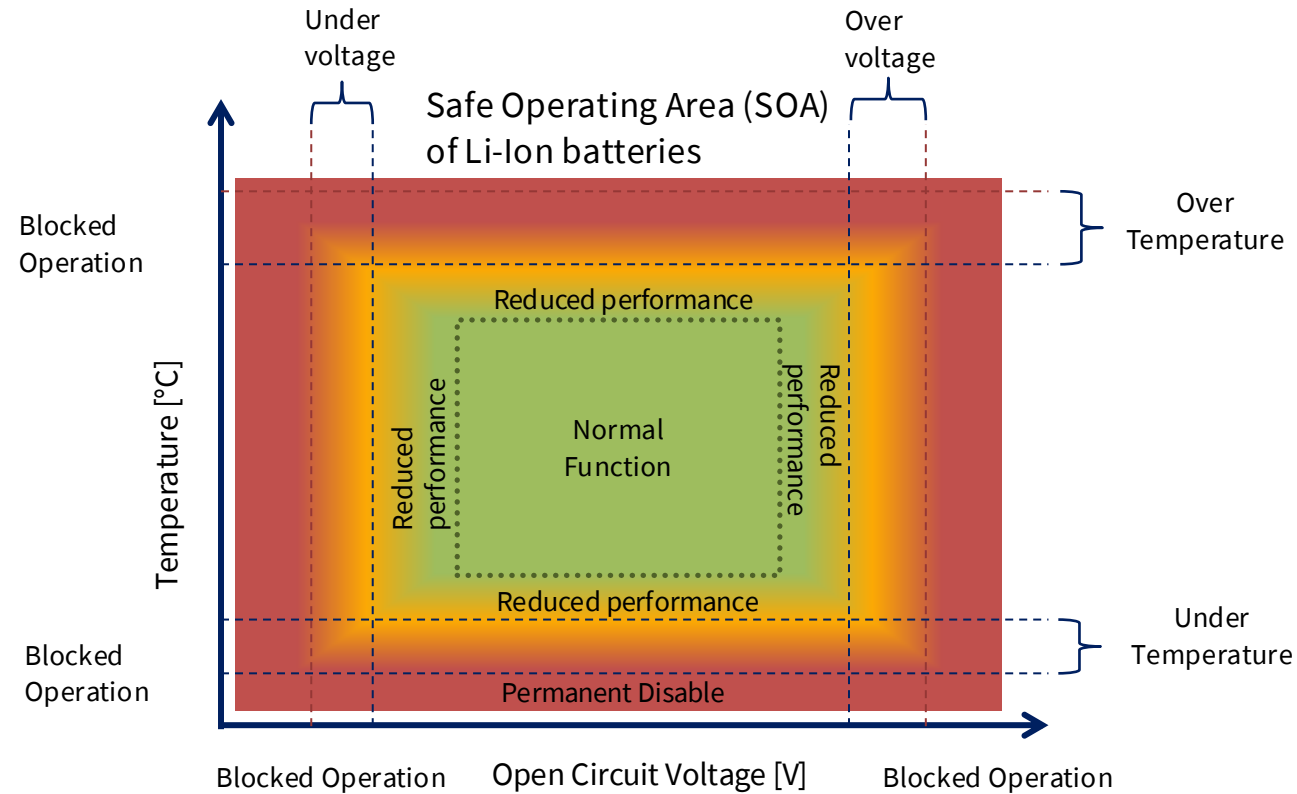
Control

- Temperature Control and Cell balancing
- Charging

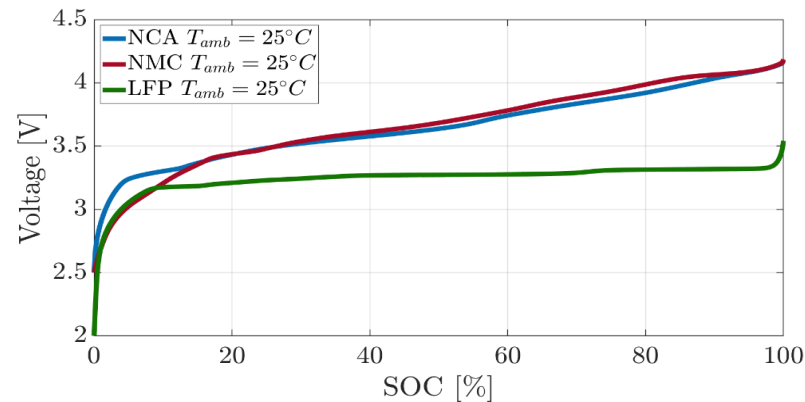
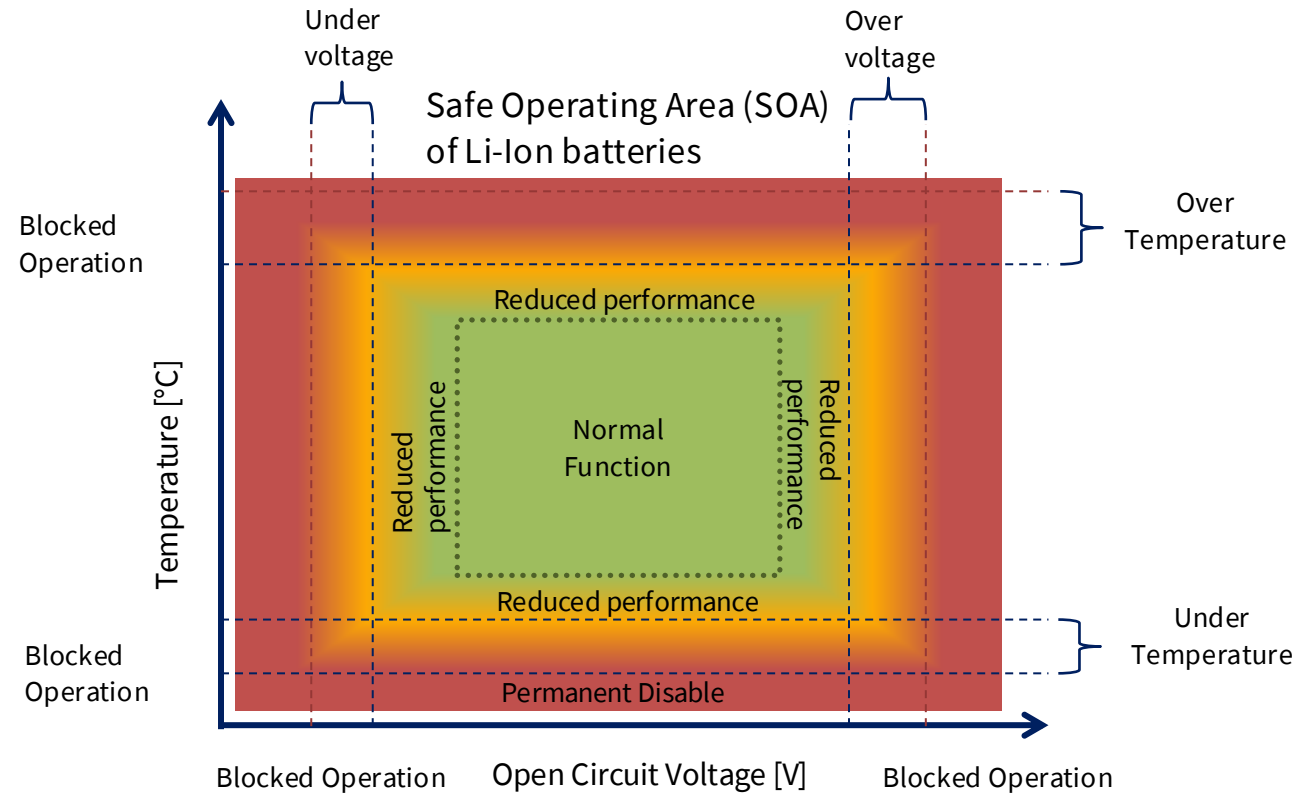
Prognostics

- Prediction of remaining useful life

Lithium-ion Battery Cell Safe Operating Area

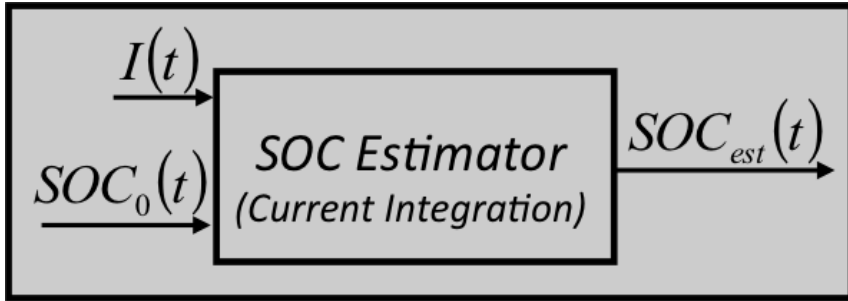


Lithium-ion Battery Cell Safe Operating Area



State of charge calculation/estimation

- The simplest method for SOC estimation is based on **Current Integration or Coulomb Counting**:



$$SOC_{est}(t) = SOC_0 - \frac{\int_{t_0}^t I(\tau) d\tau}{Q_n} \cdot 100$$

$$SOC_0 = SOC(t_0)$$

Coulomb counting method: limitations

Requires an Accurate Initial SOC

The method requires a correct starting SOC value ($\rightarrow SOC(t_0)$).

Any initial error remains and propagates over time.

There is no internal correction mechanism

$$SOC_{est}(t) = \textcolor{red}{SOC}(t_0) - \frac{\int_{t_0}^t I_{batt}(\tau) d\tau}{Q_n} \cdot 100$$

Drift Due to Current Sensor Errors

Small bias in current measurement leads to cumulative error.

Offset error integrates linearly over time.

Noise and sensor resolution contribute to long-term SOC drift.

Even a tiny current bias can produce significant SOC error over hours.

$$SOC_{est}(t) = \textcolor{black}{SOC}(t_0) - \frac{\int_{t_0}^t \textcolor{blue}{I}_{batt}(\tau) d\tau}{Q_n} \cdot 100$$

Capacity Uncertainty (Q_n is not constant)

Coulomb counting assumes known battery capacity.

In reality:

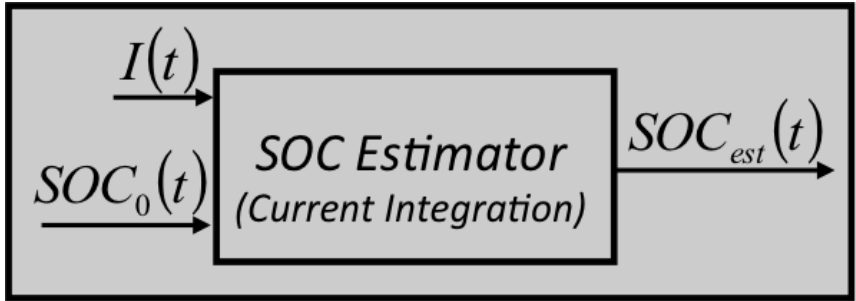
- Capacity degrades with aging.
- Capacity varies with temperature.

Incorrect capacity $\rightarrow$ systematic SOC scaling error.

$$SOC_{est}(t) = \textcolor{black}{SOC}(t_0) - \frac{\int_{t_0}^t I_{batt}(\tau) d\tau}{\textcolor{green}{Q}_n} \cdot 100$$

State of charge calculation/estimation

- The simplest method for SOC estimation is based on **Current Integration or Coulomb Counting**:



$$SOC_{est}(t) = SOC_0 - \frac{\int_{t_0}^t I(\tau) d\tau}{Q_n} \cdot 100 \quad SOC_0 = SOC(t_0)$$

- This is never in isolation though, always paired with voltage inversion (for initialization purposes)

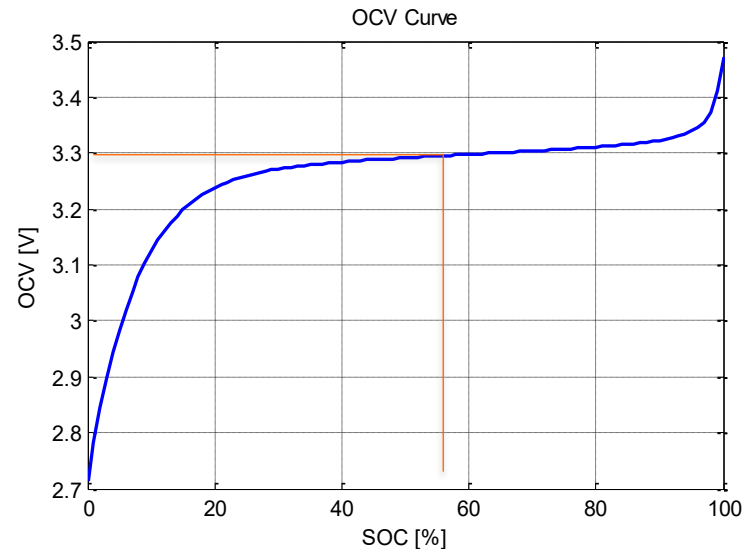
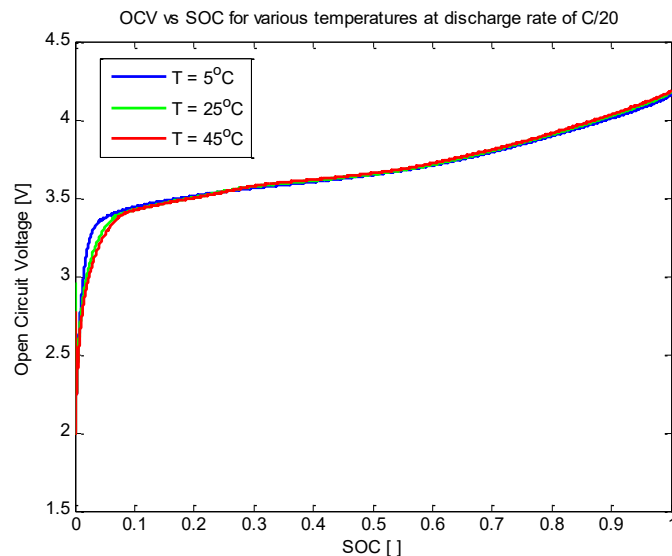
Voltage Inversion

Voltage inversion estimation determines the initial battery SOC, SOC_0 , from its Open Circuit Voltage

- It relies on the fundamental relationship:

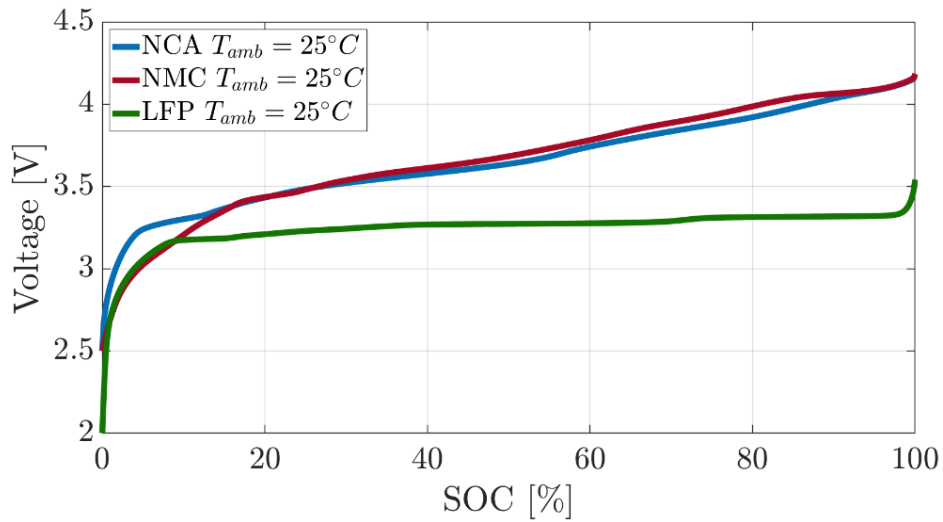
$$OCV = f(SOC)$$

- Since OCV is thermodynamically linked to lithium stoichiometry, it provides a physics-based mapping between voltage and SOC.



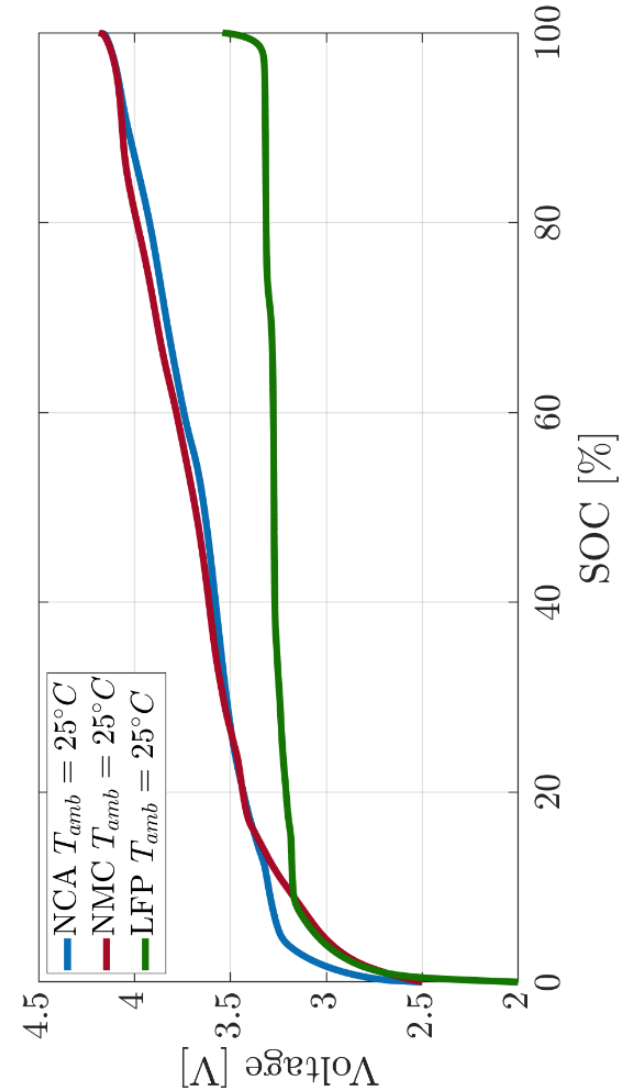
Voltage Inversion

- This approach is only valid if the open-circuit voltage can be measured, which occurs only after a rest period (zero current)
- The batteries are not at rest and approaching ‘equilibrium’ during actual vehicle operation – Relaxation can take many minutes or hours
- When the OCV curve is relatively flat with respect to SoC, the inverse relationship ($\text{SoC} = f^{-1}(\text{OCV})$) has low sensitivity and high uncertainty.



$$\text{If } \frac{dV}{d\text{SoC}} \rightarrow 0 \text{ then } \frac{d\text{SoC}}{dV} \rightarrow \infty$$

Small voltage noise $\rightarrow$ large SoC error
Poor observability (e.g., LFP plateau region)



Enhanced SOC Estimation for LFP Batteries: A Synergistic Approach Using Coulomb Counting Reset, Machine Learning, and Relaxation

Yunhong Che, Le Xu, Remus Teodorescu, Xiaosong Hu*, and Simona Onori*

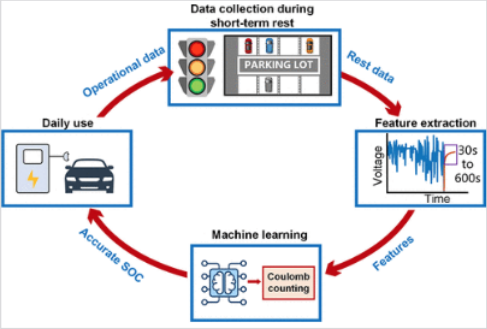
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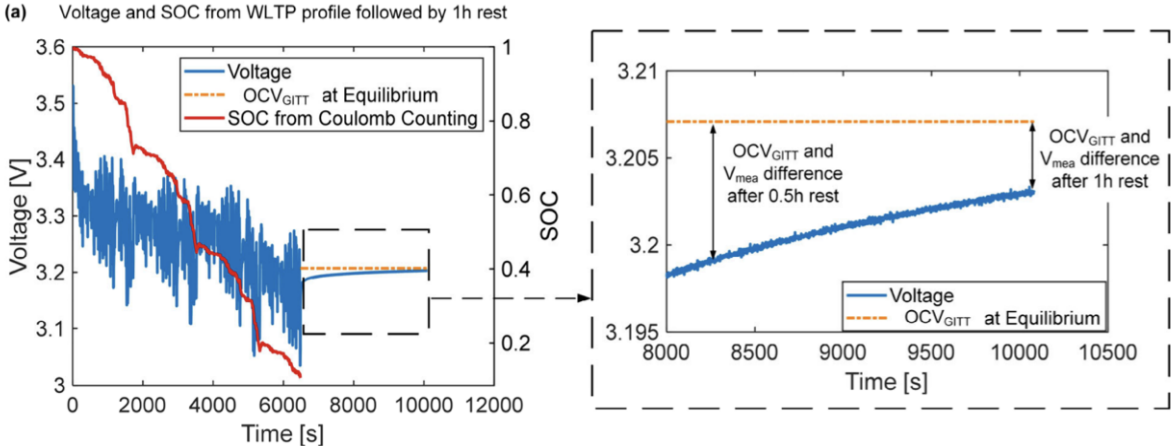
 Supporting Information (1)

Abstract

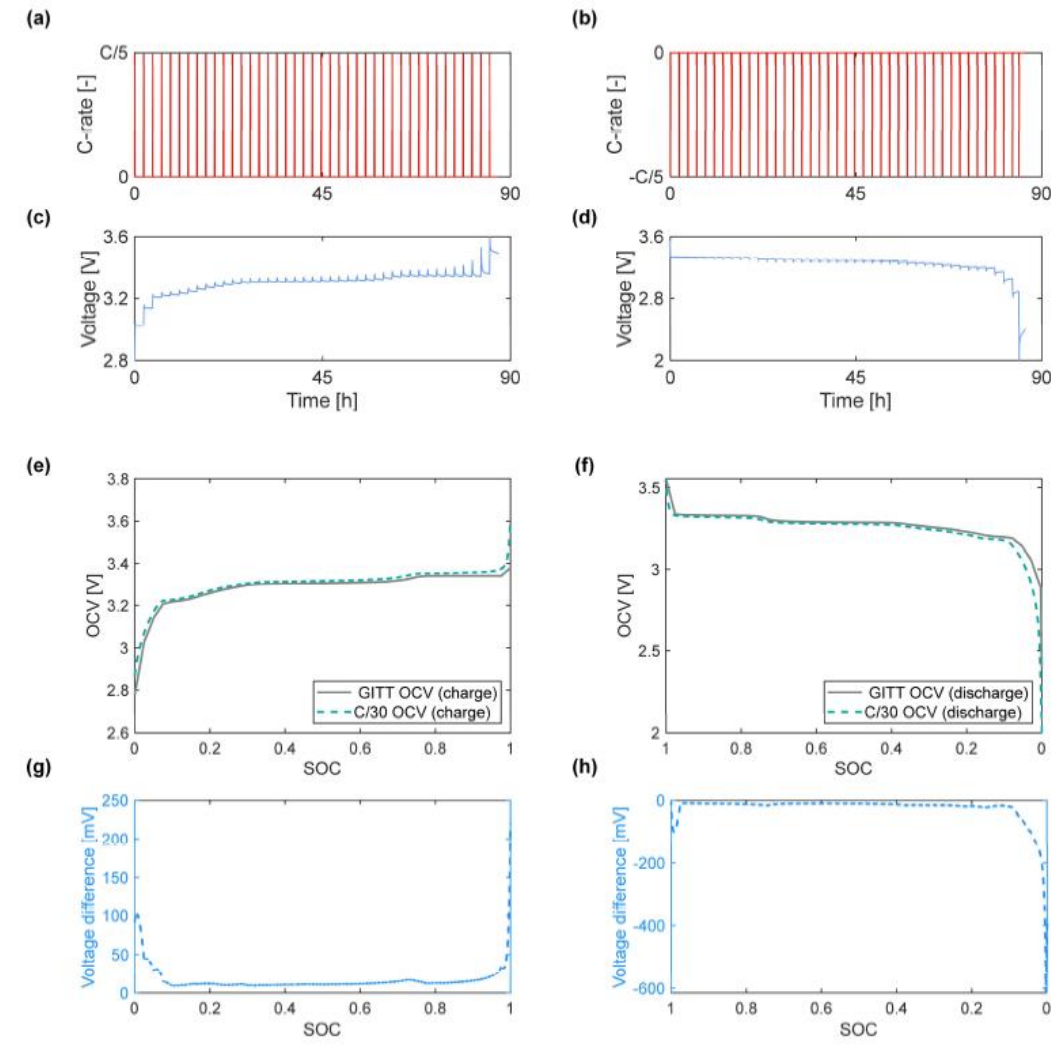
State-of-charge (SOC) estimation for lithium–iron phosphate (LFP) batteries is a challenging task due to their path-dependent behavior, flat open circuit voltage (OCV) characteristics, and hysteresis effects. This work proposes a machine-learning-based SOC estimation method designed for onboard applications, addressing the challenges of SOC initialization when using the Coulomb counting method. The proposed approach relies on low sampling frequency measurements during short-term rest periods. Experiments were conducted on LFP 26650 cells across more than 430 working conditions, involving four temperatures, three current rates, four cycling scenarios, with various resting periods at different SOC levels. A comprehensive analysis of SOC estimation errors, including initial value errors, sensor noise, and sampling frequency, is



The difference between the measured voltage V_{mea} and the OCV_{GITT} decreases as the resting time increases $\rightarrow$ long-time rest period is needed to accurately determine the battery SOC using the OCV–SOC inversion.



pseudo OCV (from C/30) and OCV obtained from GITT test



1. At equilibrium relationship:

$$V_{GITT} = f(SOC_{true}) \rightarrow \text{so the inverse map is } SOC_{true} = f^{-1}(V_{GITT})$$

2. Let the measured voltage include an offset (bias) V_{offset} :

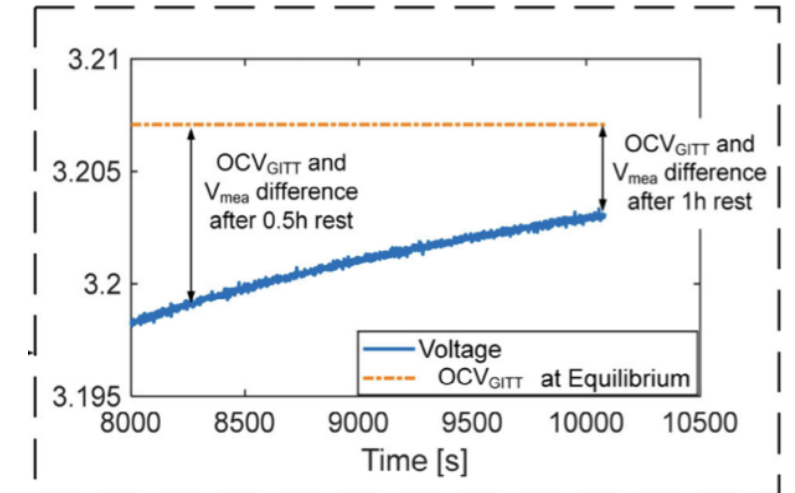
$$V_{meas} = V_{true} + V_{offset} = OCV_{GITT} + V_{offset}$$

3. The SoC estimate obtained by OCV inversion is:

$$\begin{aligned} SOC_{est} &= f^{-1}(V_{meas}) \\ &= f^{-1}(OCV_{GITT} + V_{offset}) \end{aligned}$$

4. So the SoC error is

$$\begin{aligned} \Delta SOC &\triangleq SOC_{est} - SOC_{true} \\ &= f^{-1}(OCV_{GITT} + V_{offset}) - f^{-1}(OCV_{GITT}) \end{aligned}$$



5. Linearize (first-order Taylor expansion – valid if V_{offset} is small)

→ $f^{-1}(OCV_{GITT} + V_{offset})$ around OCV_{GITT}

In the paper we use

- $V_{offset} \in (5\text{--}30 \text{ mV})$
- Compared to OCV_{GITT} scale ($\sim 2.8\text{--}3.7 \text{ V}$)

$$f^{-1}(OCV_{GITT} + V_{offset}) \approx f^{-1}(OCV_{GITT}) + \frac{df^{-1}}{dV} \big|_{OCV_{GITT}} V_{offset}$$

6. Finally, $\Delta SOC \approx \frac{df^{-1}}{dV} \big|_{OCV_{GITT}} V_{offset}$



$$\Delta SOC \approx \frac{V_{offset}}{\frac{dV}{dSOC}}$$

NOTE: $\frac{d}{dV} (f^{-1}(V)) = \frac{dSOC}{dV}$

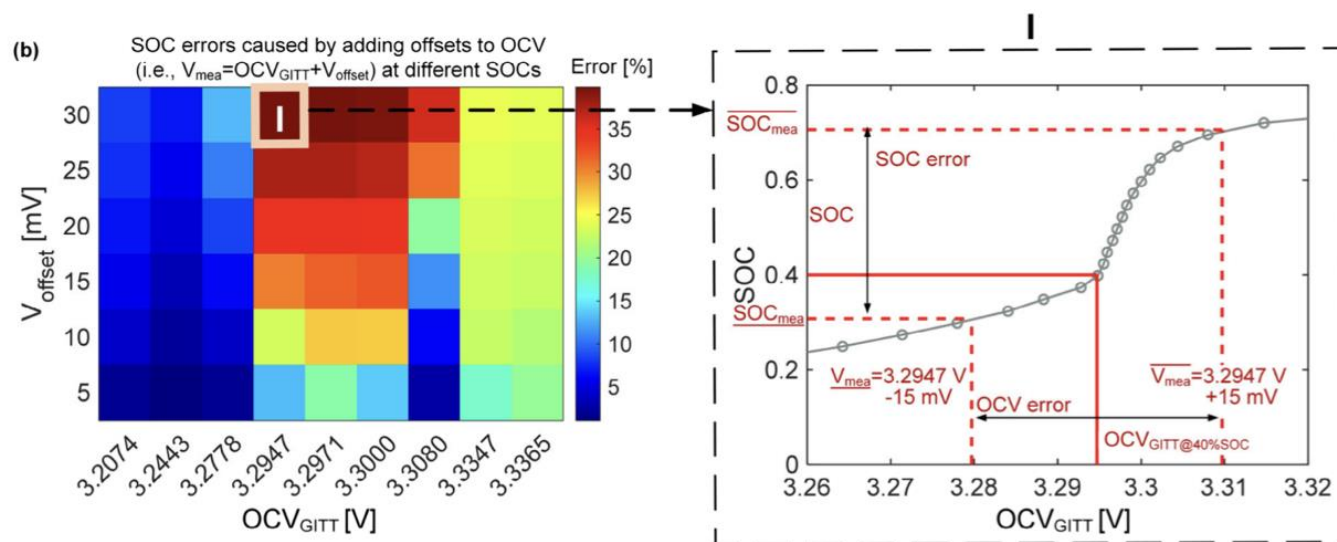
The inverse function theorem gives:

$$\frac{df^{-1}}{dV} = \frac{1}{f'(SOC)} = \frac{1}{\frac{dV}{dSOC}}$$

If the OCV curve is flat ($dV/dSOC \approx 0$), then even very small V_{offset} produces a large ΔSOC :

- Large $f'(SOC)$ → steep OCV curve → small SOC error (high sensitivity)
- Small $f'(SOC)$ → flat OCV plateau → amplified SOC error (low sensitivity)

SOC errors map caused when voltage offset noise is applied to the OCV_{GITT}



We analyze offset magnitudes ranging from 5 to 30 mV

Example:

- Consider a measured equilibrium voltage of $OCV_{GITT} = V_{true} = 3.2947 \text{ V}$ corresponding to $SOC_{true} = 40\%$.
- Introduce a $V_{offset} = 15 \text{ mV}$ offset resulting in a measured voltage of $V_{meas} = 3.3097 \text{ V}$, via OCV inversion estimated $SOC_{est} = 70.67\%$.
- Conversely, a $V_{offset} = -15 \text{ mV}$ offset yields $V_{meas} = 3.2797 \text{ V}$, corresponding to $SOC_{est} = 30.81\%$.
- $\rightarrow$ a $\pm 15 \text{ mV}$ measurement error induces a $\pm 30\%$ SOC deviation

Amperehour Throughput

vs

Capacity

vs

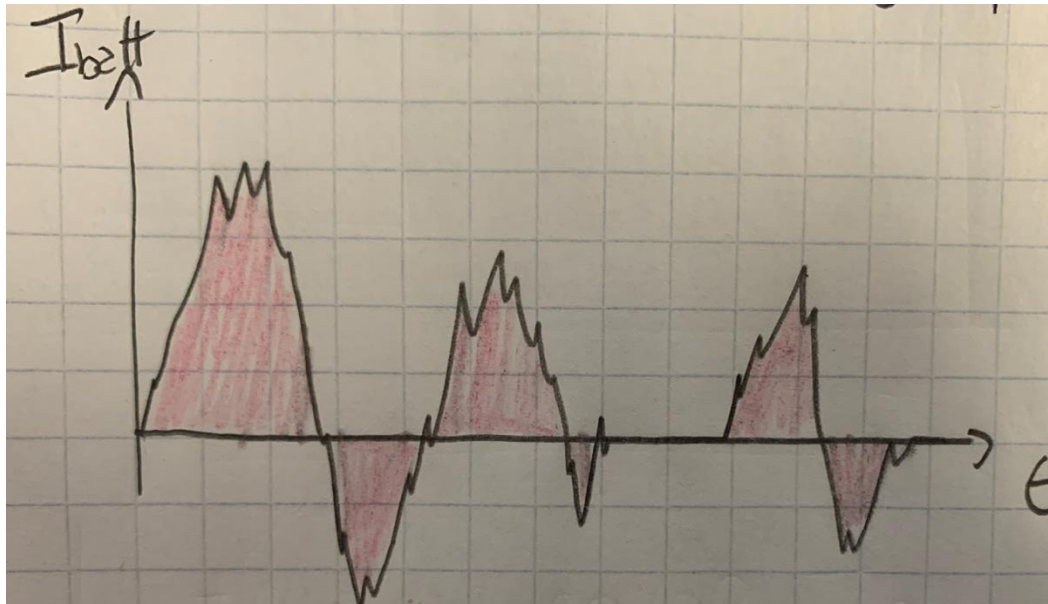
SOC

Ampere-hour Throughput

- The total charge inserted and extracted from a cell (in both charge and discharge, respectively) is expressed in terms of Ampere-hour throughput.

$$Ah_{throughput}(t) = \int_0^{t_f} \frac{|I_{batt}(t)|}{3600} dt$$

where I_{batt} is current in Amps, t is time in sec. Current convention: Positive for discharge.

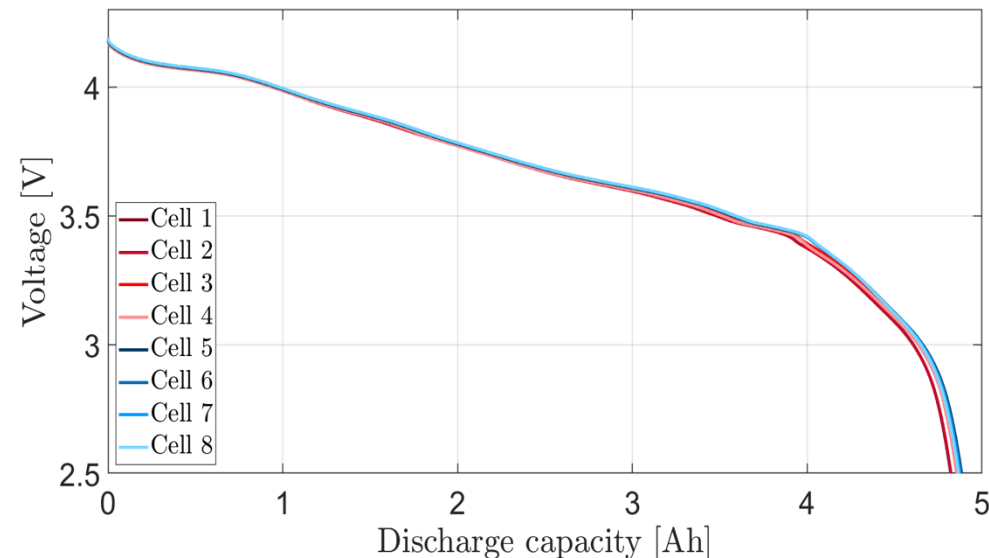


Discharge Capacity

- Discharge capacity is the charge that can be extracted from a battery during discharge:

$$Q(t) = \int_0^{t_f} \frac{I_{batt}(t)}{3600} dt$$

where I_{batt} is current in Amps, t is time in sec, $Q(t)$ is the cell capacity (discharge) in Ah.
Current convention: Positive for discharge.



State of Charge

- SOC is a dimensionless value that describes the (normalized) amount of usable charge that remains in the battery compared to the total charge capacity.

$$SOC(t) = SOC(t_0) - \frac{\int_{t_0}^t I_{batt}(\tau) d\tau}{Q_n \cdot 3600} \cdot 100$$

$$\dot{SOC}(t) = -\frac{I_{batt}}{Q_n \cdot 3600}$$

where I_{batt} is current in Amps, t is time in sec, Q_n is the cell capacity in Ah. Current convention: Positive for discharge.

Matlab Commands

- If you need the integral value at every time step (Δh), you can use the following cumulative integral command
- **Z=cumtrapz(X,Y)**: Cumulative trapezoidal numerical integration
“Z=cumtrapz(X,Y)” integrates Y with respect to the coordinates or scalar spacing specified by X. In our case, Y can be the measured current value, and X is the time vector.
cumtrapz(t,l) → gives vector over time
- If you need the integral value at the final time step (e.g, nominal capacity) you can use the following command
 - **trapz**: trapezoidal numerical integration. Syntax: trapz(X,Y)
trapz(t,l) → gives one scalar

Constant Current Profile

Discharge Capacity

- Calculated for discharge phase only

$$Q(t) = \int_0^{t_f} \frac{I(t)}{3600} dt$$

- MATLAB Implementation

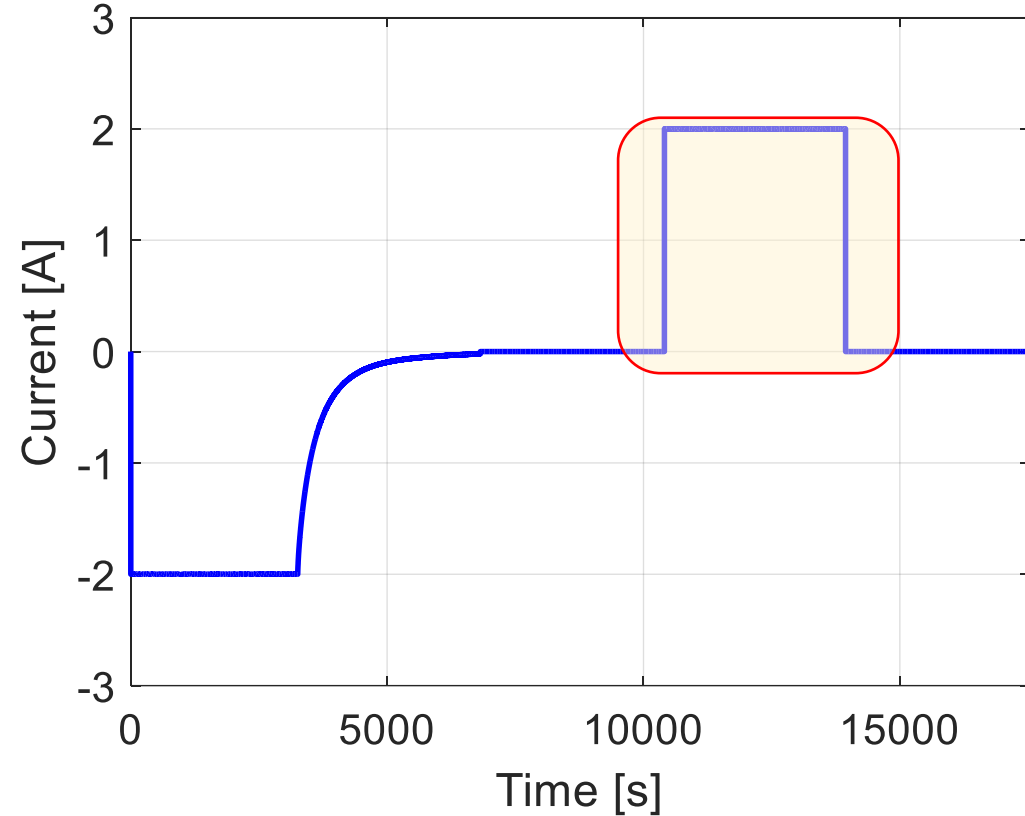
Discharge capacity as a function of time:

```
Q = cumtrapz(time,I)/3600;
```

Final discharge capacity value (nominal capacity):

```
Q = trapz(time,I)/3600;
```

= 1.96Ah



Current convention: Positive for discharge.

Constant Current Profile

State of Charge (SOC)

- Calculated for both charge and discharge phases

$$SOC(t) = \int_0^{t_f} \frac{-I(t)}{Q \times 3600} dt$$

- MATLAB Implementation:

SOC as function of time:

```
SOC = SOC_initial + cumtrapz(time, -I_data) / (Q*3600);
```

Initial SOC at time $t = 0$ can be identified from the cell terminal voltage (at rest) by using the OCV-SOC map.

US06 Current Profile

Amperehour Throughput

- Calculated for both charge and discharge phases

$$Ah_{throughput}(t) = \int_0^{t_f} \frac{|I(t)|}{3600} dt$$

- MATLAB Implementation

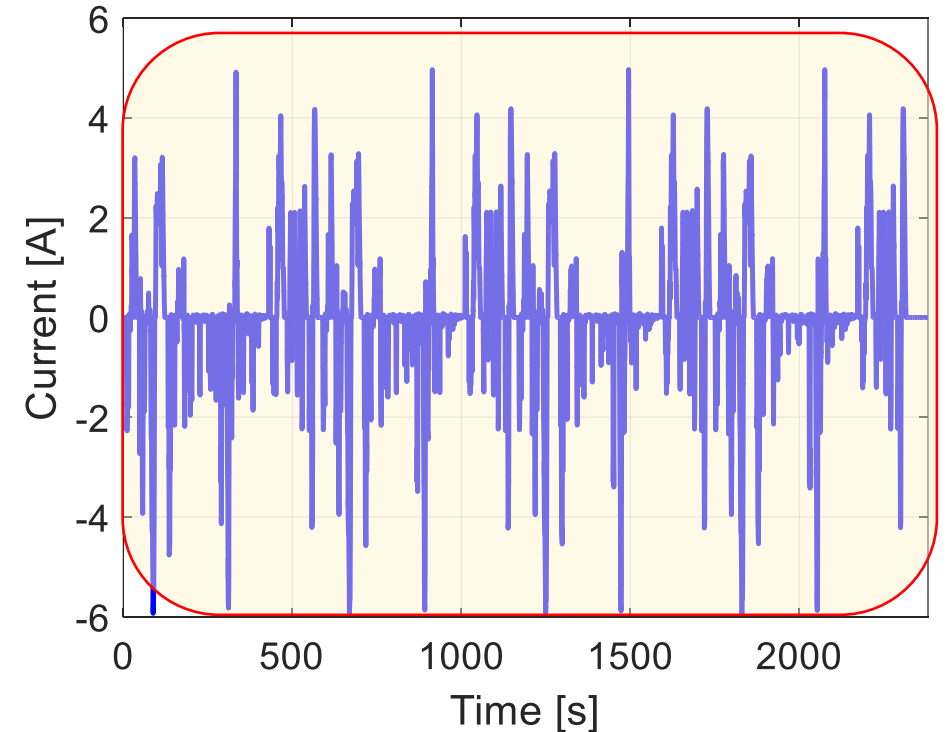
Ampere-hour throughput as a function of time
(cumulative integration at each time step)

```
Ah_throughput = cumtrapz(time,abs(I))/3600;
```

Final Ampere-hour throughput value:

```
Ah_throughput = trapz(time,abs(I))/3600;
```

= **0.4635 Ah**



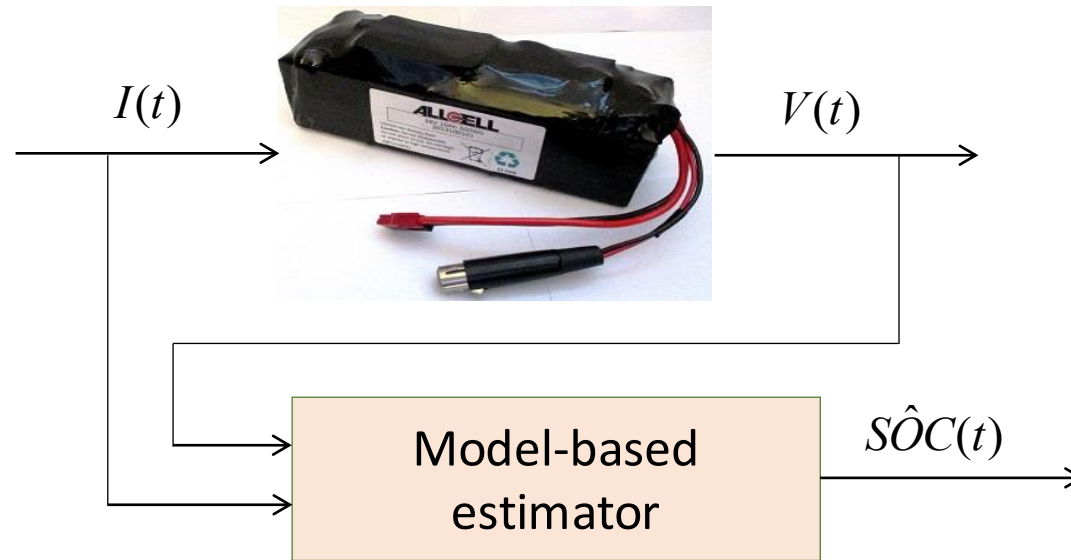
Current convention: Positive for discharge.

Discharge Capacity

- Do not calculate discharge capacity for dynamic profile.
- Compute capacity only for constant current profiles

Model-based Observer

- Observer Design: Needed to determine the internal states of an system from a given set of input/output measurements
- We use feedback to correct the mathematical model-based estimator



LTI System

Given the following system

$$\frac{dx(t)}{dt} = Ax(t) + Bu(t)$$

Its response is given by:

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} u(\tau)Bd\tau$$

and its output $y(t) = Cx(t)$ is given by:

$$y(t) = Ce^{At}x(0) + C \int_0^t e^{A(t-\tau)} u(\tau)Bd\tau$$

State response

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} u(\tau)B d\tau$$

Natural response
(Free response)

It represents the evolution of the system due only to the initial condition $x(0)$ with $u(t) = 0$

The state transition matrix $\Phi(t) = e^{At}$

- describes how the state evolves from an initial condition
- it maps the state from time 0 to time t

State response

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} u(\tau)B d\tau$$

Forced response (input response)

It represents the contribution of the input $u(t)$, assuming $x(0) = 0$

State and Output response

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} u(\tau)B d\tau$$

Total response=Natural response + Forced response

Similarly, the output response can be decomposed as ($y(t) = Cx(t)$):

$$y(t) = \underbrace{C e^{At}x(0)}_{\text{zero-input (natural) output}} + \underbrace{C \int_0^t e^{A(t-\tau)} B u(\tau) d\tau}_{\text{zero-state (forced) output}}$$

Observer

- The design of a state observer consists of introducing an auxiliary dynamical system, called an estimator, which uses available measurements and inputs to generate an estimate of the system states.
- Hypothesis → the measured output and input are noise free (no bias, no uncertainty) → Luenberger Observer

**David Luenberger**

PROFESSOR OF MANAGEMENT SCIENCE AND ENGINEERING, EMERITUS

[PRINT PROFILE](#) [EMAIL PROFILE](#) [VIEW STANFORD-ONLY PROFILE](#)

- David G. Luenberger received the B.S. degree from the California Institute of Technology and the M.S. and Ph.D. degrees from Stanford University in 1964, all in Electrical Engineering
- Since 1963 he has been on the faculty of Stanford University.
- He helped found the Department of Engineering-Economic Systems, now merged to become the Department of Management Science and Engineering, where he is currently a professor.

The Luenberger observer was introduced by David G. Luenberger in 1964 (after Kalman filter, 1960).

- D. G. Luenberger, “Observing the State of a Linear System”, *IEEE Transactions on Military Electronics*, 1964.

Observing the State of a Linear System

DAVID G. LUENBERGER, STUDENT MEMBER, IEEE

Summary—In much of modern control theory designs are based on the assumption that the state vector of the system to be controlled is available for measurement. In many practical situations only a few output quantities are available. Application of theories which assume that the state vector is known is severely limited in these cases. In this paper it is shown that the state vector of a linear system can be reconstructed from observations of the system inputs and outputs.

It is shown that the observer, which reconstructs the state vector, is itself a linear system whose complexity decreases as the number of output quantities available increases. The observer may be incorporated in the control of a system which does not have its state vector available for measurement. The observer supplies the state vector, but at the expense of adding poles to the over-all system.

techniques have been developed to find the function F for special classes of control problems. These techniques include dynamic programming [8]–[10], Pontryagin’s maximum principle [11], and methods based on Lyapunov’s theory [2], [12].

In most control situations, however, the state vector is not available for direct measurement. This means that it is not possible to evaluate the function $F[y(t), t]$. In these cases either the method must be abandoned or a reasonable substitute for the state vector must be found.

In this paper it is shown how the available system in-

Observer

- An **observer**, or **estimator**, is used to calculate state variables that are not accessible from the plant
- The observer is a model of the plant (physical system)
- **State Space Representation** (or also Input-State-Output) – (LTI= Linear time invariant system)

$$\begin{aligned}\frac{dx(t)}{dt} &= Ax(t) + Bu(t) \\ y(t) &= Cx(t)\end{aligned}$$

where

- $x(t) \in \mathbb{R}^n$ (state vector $\rightarrow$ **system order** n)) (# number of states = n)
- $u(t) \in \mathbb{R}^m$ (input vector) (# inputs = m)
- $y(t) \in \mathbb{R}^p$ (output vector) (# outputs = p)

with matrices $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$

FACT: We measure $y(t)$ directly through sensors, but we cannot measure the state vector $x(t)$.

GOAL: We want to try to reconstruct (observe) the state $x(t)$ from available measurements (and inputs), i.e. build a state observer.

Observability

Observability is about the ability to estimate the internal states from measured input/output data.

For a system

$$\begin{aligned}\dot{x} &= Ax + Bu, \\ y &= Cx + Du\end{aligned}$$

Observability means:

- Can we uniquely reconstruct the initial state $x(0)$ from knowledge of $u(t)$ and $y(t)$ over a finite time interval?
- Observability depends on whether different initial states produce distinguishable outputs.

A LTI system is **observable** if given

$$u(t), \quad y(t) \quad t \in [0, T]$$

we can uniquely determine any initial state $x(0)$ from the input and output history over a finite time interval T .

- **Unobservable:** there are indistinguishable initial states.

Observability

The system is **observable** if:

The initial state $x(0)$ can be uniquely determined from knowledge of the input $u(t)$ and output $y(t)$ over a finite time interval.

$$y(t) = Ce^{At}x(0) + C \int_0^t e^{A(t-\tau)} u(\tau) B d\tau$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)} u(\tau) B d\tau$$

→ If we know $x(0)$, then we know $x(t)$!

Observability

- Two initial states $x_1(0)$ and $x_2(0)$ are **indistinguishable** (for zero input) if and only if, given the natural responses:

$$\begin{aligned}y_1 &= Ce^{At}x_1(0) \\ y_2 &= Ce^{At}x_2(0)\end{aligned}$$

$$y_1 = y_2 \quad \longrightarrow \quad Ce^{At}x_1(0) = Ce^{At}x_2(0) \quad \forall t \geq 0$$

Equivalently,

$$Ce^{At}(x_1(0) - x_2(0)) = 0 \quad \forall t \geq 0$$

- We cannot distinguish whether the natural output response was generated by $x_1(0)$ or $x_2(0)$, because they produce identical outputs for all time.

Observability

- How do we check this condition algebraically? If a state $x_0 = x(0)$ is observable (distinguishable) how can we calculate it from the plant model, input and output?
- We can derive the observability matrix *directly* from the natural output response (zero input). The initial states appear only in the natural response

- Assume $u(t) = 0$. Then. $x(t) = e^{At}x(0)$, $y(t) = Cx(t) = Ce^{At}x(0)$

- Now we look at $y(t)$ and its time-derivatives at $t = 0$ (Since the output $y(t)$ is known (and so its derivatives)) $y(0) = Cx(0)$

- Calculate the first derivative:

$$\dot{y}(t) = \frac{d}{dt}(Ce^{At}x(0)) = CAe^{At}x(0) \Rightarrow$$
$$\dot{y}(0) = CAx(0)$$

Observability

Differentiate again...

$$\ddot{y}(t) = CA^2 e^{At} x(0) \Rightarrow \\ \ddot{y}(0) = CA^2 x(0)$$

... until $n-1$

$$y^{(n-1)}(0) = CA^{n-1} x(0)$$

Stack equations:

$$\begin{bmatrix} y(0) \\ \dot{y}(0) \\ \ddot{y}(0) \\ \vdots \\ y^{(n-1)}(0) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} x(0)$$

Observability matrix is:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

A LTI system is observable if and only if the observability matrix has full rank n .

Observability

Observability test

System is observable $\Leftrightarrow \text{rank}(\mathcal{O}) = n$

Open-loop versus Closed-loop

Luenberger observer

